

Technology Stack

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML, CSS, JavaScript) <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, JavaScript technology]</i>	<i>Simple, lightweight form-based UI that works in any browser without extra frameworks</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Flask (Python) technology]</i>	<i>Lightweight Python web framework, easy to integrate with ML models</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV file (flat file storage) technology]</i>	<i>Simple and sufficient for a static training dataset in this project scope mapping)]</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Cloud (planned) technology]</i>	<i>Scalable, reliable hosting suited for future public deployment cost-effective hosting)]</i>
5	Version Control & CI/CD	<i>Git & GitHub technology]</i>	<i>Industry-standard version control for tracking changes and collaboration</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, XGBoost, Pandas, NumPy technology]</i>	<i>Core ML libraries used for model training, comparison, and data preprocessing</i>